Pizza Sales SQL Analysis

COMPLETE QUERY DOCUMENTATION & PROBLEM STATEMENTS

BASIC ANALYSIS

1. Retrieve the total number of orders placed.

```sql
SELECT COUNT(order_id) AS total_orders FROM orders;
```

2. Calculate the total revenue generated from pizza sales.

```sql
SELECT SUM(order_details.quantity * pizzas.price) AS total_sales
FROM order_details
JOIN pizzas ON pizzas.pizza_id = order_details.pizza_id;
```

3. Identify the highest-priced pizza.

```sql
SELECT pizza_types.name, pizzas.price
FROM pizza_types
JOIN pizzas ON pizza_types.pizza_type_id = pizzas.pizza_type_id
ORDER BY pizzas.price DESC LIMIT 1;
```

4. Identify the most common pizza size ordered.

```sql
SELECT pizzas.size, COUNT(order_details.order_details_id) AS order_count
FROM pizzas
JOIN order_details ON pizzas.pizza_id = order_details.pizza_id
GROUP BY pizzas.size ORDER BY order_count DESC;
```

5. List the top 5 most ordered pizza types along with their quantities.

```sql
SELECT pizza_types.name, SUM(order_details.quantity) AS quantity
FROM pizza_types
JOIN pizzas ON pizza_types.pizza_type_id = pizzas.pizza_type_id
JOIN order_details ON order_details.pizza_id = pizzas.pizza_id
GROUP BY pizza_types.name ORDER BY quantity DESC LIMIT 5;
```

INTERMEDIATE ANALYSIS

6. Join the necessary tables to find the total quantity of each pizza category ordered.

```sql
SELECT pizza_types.category, SUM(order_details.quantity) AS quantity
FROM pizza_types
JOIN pizzas ON pizza_types.pizza_type_id = pizzas.pizza_type_id
JOIN order_details ON order_details.pizza_id = pizzas.pizza_id
GROUP BY pizza_types.category ORDER BY quantity DESC;
```

7. Determine the distribution of orders by hour of the day.

```sql
SELECT HOUR(order_time) AS hour, COUNT(order_id) AS order_count
FROM orders GROUP BY HOUR(order_time);
```

8. Join relevant tables to find the category-wise distribution of pizzas.

```sql
SELECT category, COUNT(name) FROM pizza_types GROUP BY category;
```

9. Group the orders by date and calculate the average number of pizzas ordered per day.

```sql
SELECT ROUND(AVG(quantity), 0) AS avg_pizza_order_per_day
FROM (
    SELECT orders.order_date, SUM(order_details.quantity) AS quantity
    FROM orders JOIN order_details ON orders.order_id = order_details.order_id
    GROUP BY orders.order_date
) AS order_quantity;
```

10. Determine the top 3 most ordered pizza types based on revenue.

```sql
SELECT pizza_types.name, SUM(order_details.quantity * pizzas.price) AS revenue
FROM pizza_types
JOIN pizzas ON pizzas.pizza_type_id = pizza_types.pizza_type_id
JOIN order_details ON order_details.pizza_id = pizzas.pizza_id
GROUP BY pizza_types.name ORDER BY revenue DESC LIMIT 3;
```

ADVANCED ANALYSIS

11. Calculate the percentage contribution of each pizza type to total revenue.

```sql
SELECT pizza_types.category,
    (SUM(order_details.quantity * pizzas.price) / (
        SELECT SUM(order_details.quantity * pizzas.price) AS total_sales
        FROM order_details JOIN pizzas ON pizzas.pizza_id = order_details.pizza_id
    )) * 100 AS revenue
FROM pizza_types
JOIN pizzas ON pizzas.pizza_type_id = pizza_types.pizza_type_id
JOIN order_details ON order_details.pizza_id = pizzas.pizza_id
GROUP BY pizza_types.category ORDER BY revenue DESC;
```

12. Analyze the cumulative revenue generated over time.

```sql
SELECT order_date, SUM(revenue) OVER(ORDER BY order_date) AS cum_revenue
FROM (
    SELECT orders.order_date, SUM(order_details.quantity * pizzas.price) AS revenue
    FROM order_details
    JOIN pizzas ON order_details.pizza_id = pizzas.pizza_id
    JOIN orders ON orders.order_id = order_details.order_id
    GROUP BY orders.order_date
) AS sales;
```

13. Determine the top 3 most ordered pizza types based on revenue for each pizza category.

```sql
SELECT category, name, revenue FROM (
    SELECT category, name, revenue,
           RANK() OVER (PARTITION BY category ORDER BY revenue DESC) AS rn
    FROM (
        SELECT pizza_types.category, pizza_types.name,
               SUM(order_details.quantity * pizzas.price) AS revenue
        FROM pizza_types
        JOIN pizzas ON pizza_types.pizza_type_id = pizzas.pizza_type_id
        JOIN order_details ON order_details.pizza_id = pizzas.pizza_id
        GROUP BY pizza_types.category, pizza_types.name
    ) AS a
) AS b WHERE rn <= 3;
```